NGSS and 5E Planning Template

Grade:	**Lesson/Unit Topic:**	
Brief Description of the main idea of the lesson plan and content area:	Bones and Muscles of an Elephant.	
Performance Expectation (this should be the NGSS or SD standard)**:**	**4.LS1-1** Construct an argument that plants and animals have internal and external structures that function to support survival, growth, behavior, and reproduction. https://www.nextgenscience.org/dci-arrangement/4-ls1-molecules-organisms-structures-and-processes	
Specific Learning Outcomes (what will students be able to DO after this lesson/unit?**)** Please put this in student-friendly terms (I can statement)	Students will be able to: "I can…" *Construct an explanation *identify the bones and muscles of an elephant *describe the functions served by the bones and muscles of an elephant	
Prior Student Knowledge (when have they learned about this topic before- in or out of school):	What do you know about bones and muscles? What happens if you break a bone or pull a muscle?	
Possible Preconceptions or Misconceptions	Students may think bones are completely rigid with no flex at all. In actuality, bones, like fresh sticks and even steel beams, will grow and shrink with the weather and time of day, and when a bone is pressed, it flexes before it breaks. Bones like the backbone, or spine, are made up of many smaller parts called vertebrae, which allow it to be more flexible. In fact, the inside of all bones are spongy, not hard. Students may think muscles grow, perhaps like balloons, when one works out or becomes stronger, when in fact they tear and rebuild themselves to become "stronger". Additionally there are many types of muscles Cartilage is another very important part of the body, but that's for a different lesson perhaps. Though students might think there's only muscle/bone?	
Science and Engineering Practices Developing and Using Mo…	Crosscutting Concepts Systems and Systems Models	Disciplinary Core Ideas LS1.A Life Science: Structure and F…
What will the students DO? (add language below:)	What will the students think about?(add language below:)	What is the science topic being investigated? (add language below:)

Lesson Plan- 5E Model

Engage: Short activity to promote curiosity and elicit current knowledge. Include your anchoring phenomenon, how it will be used and why it will get students thinking or wondering. Link any resources and include an explanation of why they are included, how they will be used, and why they are important. Add all questions that will be asked of the students and what questions you would expect the students to ask.

Use the picture from the textbook, page 83, and say something like "here is the elephant's skeleton and muscles". Ask probing questions "what does this do", etc to each of the things listed on the page

Explore:
Students are engaged in an exploration activity to provide them with a common base. This should be a hands-on/minds-on activity. Include materials needed, the purpose of the activity, how the activity will be run, questions that will be posed to students, and expected outcomes or questions the students will have. This is a good place for students to Gather, Reason, and Communicate. Students may be using their Science and Engineering Practice here and their science notebooks to note understandings and questions. Phenomena should be considered here.

Here is a 3d model of an elephant skeleton, take some time to explore it, and ask yourself "what does this bone do"
Teacher should roam and make sure students are on topic, give them 10 ish minutes

Explain: Students focus on their engage and explore activities and use additional resources to build an understanding of the concepts. This is where students will dive into the crosscutting concepts to think about what they are learning. Students should use their science notebooks to add information and make connections. This is a good place for a picture book, additional book or text resources, videos, and vetted web sources. Revisit the phenomena here.

Teacher asks probing questions; Students explain

"What do all of the bones of an elephant make up? Which part of the body is made up of many small bones? Which bones make up the head? Which bone runs the length of the back? Which bones surround the heart and lungs? What bone structure is at the top of the rear legs? What function does the spongy part of the bone serve? What happens when a skeletal muscle contracts? What do the ribs, backbone, and skull have in common? What do the pelvis, backbone and jawbone have in common?"
One at a time, remember to give a couple seconds after asking before taking answers

Extend: Students apply their science understanding in a new way or with an additional activity. This should challenge students to think about their learning with a deeper or more broad understanding. Revisit the phenomena here.

"There's multiple types of muscles; the ones you think of when you think of muscle is the skeletal muscle..
Do you think an elephant would have a better chance to survive with a broken rib or a broken jaw?"

Revisit the possible misconceptions from earlier and maybe ask the students what they think other misconceptions could be?
show a picture of the skeleton of a dog, how does this compare to an elephant?; probing question

Evaluate: Students should be able to meet the three dimensions of the NGSS or SD science standard (Performance expectation.) Note where in the lesson will the students be formatively assessed and what their summative assessment will look like. Reminder, the summative assessment should align with the wording for the standard.

Summative assessment: Description of action students will display to show understanding and proficiency in the standard. (Full understanding or proficiency and evidence of that proficiency)	Okay in your science notebooks, i'd like for you to write down the answers to these questions as an exit ticket "What structure protects the brain" memorization question "How do bones and muscles work together" explain "Comapred to other land animals, the bones of an elephant are very think and heavy. How might this help an elephant survive" infer
What would limited or partial understanding look like?	Im not looking at memorization, i am much more interested in explain and infer, does the student understand the concepts or are they simply repeating what the textbook or teacher said verbatim
Additional resources or links	https://sketchfab.com/3d-models/african-elephant-skeleton-0a7cb290616442c88f89107d9a11f8f0
AI-generated information	n/a

links	
What human reviews or modifications were made to AI output?	

	2: Proficient	1: Nearing Proficiency	0: not yet
Make an explanation	I made an explanation that is present and well done	I made an explanation	I did not make an explanation
Identify	I am able to identify the bones and muscles	I am able to identify some of the bones and muscles	I did not identify
Describe	I can describe the purpose of bones and muscles	I can describe the purpose of bones and muscles with some accuracy	I did not describe